



# Persongify - Image to Music Generation

*Theme:* Software Engineering

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## **Motivation**

There were two recent moments during our explorations with AI that made us pause with fascination and awe. The first was when we learned about the website “They See Your Photos” and saw the in depth textual summary that could easily be generated from an image. The second was when our friend group spent an evening using AI to generate songs about each other, and the results were so catchy that we continue to listen to them now.

Our project idea is to combine these two moments of awe into a single, mind bending application that can take a photo as an input and generate a catchy, relevant, and eerily accurate song about it.

## **Problem Statement**

We want to combine two deep learning methods into one, continuous pipeline. We plan to take images as input and produce novel songs as an output. We do not intend on writing custom models, but rather plan on leveraging existing APIs from Google to create a seamless application. The goal of our project is to create an engaging application that is entertaining and inspires interest in Deep Learning algorithms.

**Main Tasks** - All tasks will be completed together, either using paired coding or asynchronously with code reviews.

- **Image Summary Implementation:** Develop a python application that accepts an input image and queries an API (likely Google Vision API) to get a text summary.
- **Text to Song Implementation:** Develop a python implementation that takes an input text and queries an API (likely Google Lyra API) to get a generated song.
- **Combined Pipeline:** Take an image as an input, get the summary and reformat it as an input to the song generation model. At this point, we hope to have a full working pipeline for image → song.
- **Front End and Application Polishing:** Develop a webserver with an attractive frontend to make pipeline use easy and smooth.